

```

/*
CIS 180
Assignment 14
Agarwal, Rishita
*/

#include <iostream>
#include <string>
using namespace std;

// Declaration of Number class
class Number
{
private:
    int number;

public:
    static string lessThan20[20];
    static string tens[10];
    static string hundred;
    static string thousand;

    // Constructor
    Number(int x)
    {
        number = x;
    }

    // Print function
    void print();
};

// Static member definitions
string Number::lessThan20[20] =
{
    "zero", "one", "two", "three", "four",
    "five", "six", "seven", "eight", "nine",
    "ten", "eleven", "twelve", "thirteen",
    "fourteen", "fifteen", "sixteen",
    "seventeen", "eighteen", "nineteen"
};

string Number::tens[10] =
{

```

```

    "", "", "twenty", "thirty", "forty",
    "fifty", "sixty", "seventy",
    "eighty", "ninety"
};

string Number::hundred = "hundred";
string Number::thousand = "thousand";

// Print function
void Number::print()
{
    int remainder = number;

    if (remainder < 0 || remainder > 9999)
    {
        cout << "Invalid input." << endl << endl;
        return;
    }

    cout << "The English description is:";

    if (remainder == 0)
    {
        cout << " zero";
    }
    else
    {
        // Thousands
        int n_thousands = remainder / 1000;
        remainder %= 1000;

        if (n_thousands > 0)
        {
            cout << " " << lessThan20[n_thousands]
                << " " << thousand;
        }

        // Hundreds
        int n_hundreds = remainder / 100;
        remainder %= 100;

        if (n_hundreds > 0)
        {
            cout << " " << lessThan20[n_hundreds]

```

```

        << " " << hundred;
    }

    // Add "and" if needed
    if ((n_thousands > 0 || n_hundreds > 0) && remainder > 0)
    {
        cout << " and";
    }

    // Tens
    if (remainder >= 20)
    {
        int n_tens = remainder / 10;
        remainder %= 10;

        cout << " " << tens[n_tens];
    }

    // Ones / Teens
    if (remainder > 0 && remainder < 20)
    {
        cout << " " << lessThan20[remainder];
    }
}

cout << endl << endl;
}

int main()
{
    int number;

    cout << "This program translates whole dollar amounts\n";
    cout << "in the range 0 through 9999 into an English\n";
    cout << "description of the number.\n\n";
    cout << "Enter -1 to end input.\n\n";

    do
    {
        cout << "Enter a whole dollar amount: ";
        cin >> number;

        if (number == -1)
            break;
    }
}

```

```
    Number n(number);  
    n.print();  
  
} while (true);  
  
return 0;  
}
```